Newton 法

求 $f(x)=0$ 的根 $\quad x_{k+1} = x_k - \alpha \dfrac{f(x_k)}{f'(x_k)}$

$\min f(x)$ 即求 $\nabla f(x) = 0$

$$x_{k+1} = x_k - \alpha [\mathrm{Hess} f(x_k)]^{-1} \nabla f(x_k)$$

收敛性证明:

$$x^{k+1} - x^* = x^k - x^* - \eta [\mathrm{Hess} f_k]^{-1} \nabla f_k$$

$$= [\mathrm{Hess} f_k]^{-1} [\mathrm{Hess} f_k \cdot (x^k - x^*) - \eta \nabla f_k]$$

Since $\nabla f^* = 0$, apply Tylor expand

$$\nabla f_k = \nabla f_k - \nabla f^*$$

$$= \int_0^1 \mathrm{Hess} f(x^k + t(x^* - x^k)) \cdot (x^* - x^k)\, dt$$

We have

$$\| \mathrm{Hess} f_k (x^k - x^*) - \eta \nabla f_k \|$$

$$= \| \int_0^1 [\mathrm{Hess} f_k - \eta \mathrm{Hess} f(x^k + t(x^* - x^k))] \cdot (x^* - x^k)\, dt \|$$

$$\leq \int_0^1 \| \mathrm{Hess} f_k - \eta \mathrm{Hess} f(x^k + t(x^* - x^k)) \| \cdot \| x^* - x^k \|\, dt$$

$$\leq \| x^* - x^k \|^2 \int_0^1 L t\, dt$$

$$= \frac{1}{2} L \| x^* - x^k \|^2$$

Since $\mathrm{Hess} f^*$ nonsingler

$$\exists r \ \forall x^k \in B_r(x^*), \quad \| (\mathrm{Hess} f_k)^{-1} \| \leq 2 \| (\mathrm{Hess} f^*)^{-1} \|$$

So

$$\| x^{k+1} - x^* \| \leq L \| (\mathrm{Hess} f^*)^{-1} \| \cdot \| x^* - x^k \|^2 = \tilde{L} \| x^* - x^k \|^2$$

in which $\tilde{L} = L \| (\mathrm{Hess} f^*)^{-1} \|$

choosing $x_0 \in B_{\min(r, \frac{1}{2}\tilde{L})}(x^*)$

$\{x^k\}$ convergence to x^*, and rate of convergence is quadratic.

Quasi-Newton BFGS

Step 1. Estimate an initial design $\mathbf{x}^{(0)}$. Choose a symmetric positive definite $n \times n$ matrix $\mathbf{H}^{(0)}$ as an estimate for the Hessian of the cost function. In the absence of more information, let $\mathbf{H}^{(0)} = \mathbf{I}$. Choose a convergence parameter ε. Set $k = 0$, and compute the gradient vector as $\mathbf{c}^{(0)} = \nabla f(\mathbf{x}^{(0)})$.
Step 2. Calculate the norm of the gradient vector as $||\mathbf{c}^{(k)}||$. If $||\mathbf{c}^{(k)}|| < \varepsilon$, stop the iterative process; otherwise, continue.
Step 3. Solve the following linear system of equations to obtain the search direction:

$$\mathbf{H}^{(k)}\mathbf{d}^{(k)} = -\mathbf{c}^{(k)} \tag{11.40}$$

Step 4. Compute the optimum step size:

$$\alpha_k = \alpha \text{ to minimize } f(\mathbf{x}^{(k)} + \alpha\mathbf{d}^{(k)}) \tag{11.41}$$

Step 5. Update the design as

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k\mathbf{d}^{(k)} \tag{11.42}$$

Step 6. Update the Hessian approximation for the cost function as

$$\mathbf{H}^{(k+1)} = \mathbf{H}^{(k)} + \mathbf{D}^{(k)} + \mathbf{E}^{(k)} \tag{11.43}$$

where the correction matrices $\mathbf{D}^{(k)}$ and $\mathbf{E}^{(k)}$ are given as

$$\mathbf{D}^{(k)} = \frac{\mathbf{y}^{(k)}\mathbf{y}^{(k)T}}{(\mathbf{y}^{(k)} \cdot \mathbf{s}^{(k)})}; \quad \mathbf{E}^{(k)} = \frac{\mathbf{c}^{(k)}\mathbf{c}^{(k)T}}{(\mathbf{c}^{(k)} \cdot \mathbf{d}^{(k)})} \tag{11.44}$$

$$\mathbf{s}^{(k)} = a_k\mathbf{d}^{(k)} \text{(change in design)}; \quad \mathbf{y}^{(k)} = \mathbf{c}^{(k+1)} - \mathbf{c}^{(k)} \text{ (change in gradient)}; \quad \mathbf{c}^{(k+1)} = \nabla f(\mathbf{x}^{(k+1)}) \tag{11.45}$$

Step 7. Set $k = k + 1$ and go to Step 2.